

Finland

Sovereign government data centre network — capacity, legal posture, provider landscape and migration path.

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1 Starting point

Population	5.64 m
GDP	EUR 282 bn
Public administration (NACE O)	116 k
Electricity price	74.8 EUR/MWh
Renewables	54.3%
Live hyperscaler regions	1

2 What is structurally different

- Frontline exposure. A land border with Russia or Belarus (or a Black Sea coast facing the war) changes the threat model from *geopolitical supply disruption* to *kinetic and sabotage risk against the facilities themselves*. Defense and security workloads are scaled up 1.5x/1.25x in the baseline, and at least one site should be hardened (EMP/blast, autonomous power for weeks, not hours). A purely national footprint cannot provide the out-of-country cold copy that Estonia's Data Embassy already demonstrates; this is the first item to revisit when the EU federation layer (Dutch GOAL.md section 16) is modelled.
- Cheap, clean power (75 EUR/MWh, 54% renewables). The economics favour building larger sites than the 12 MW planning unit and offering spare sovereign capacity to partners - a reason to revisit the site-size assumption.

3 Capacity

Servers	1,715 (CPU 1,173 / GPU 89 / storage 453)
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IT critical load	2.9 MW
Facility design load	4.3 MW
Sites	3 (by capacity 1, floor 3)
Average per site	1.4 MW
Site count set by	the minimum-sites floor, not capacity
CAPEX	EUR 104 m
OPEX	EUR 7 m / yr

4 Proposed geography

Region	Role	Share	MW	Notes
Helsinki - Uusimaa	Primary civil cloud	41%	1.8	Valtori estate, FICIX, C-Lion1 landing; 30 min from Tallinn.
Tampere - Pirkanmaa	Sovereign secondary	27%	1.2	Inland, 180 km separation, strong grid.
Oulu / North	Government / continuity	23%	1.0	Cheapest power, free cooling, far from the frontier; adds ~10 ms.
Kajaani - Kuopio	Strategic reserve	10%	0.4	LUMI (Kajaani) proves power and cooling for large loads.

First-pass geographic hypotheses encoding only the obvious constraints, to be replaced by scored site selection.

5 Legal and regulatory posture

Governing instrument	Act on Information Management in Public Administration (906/2019)
Cloud certification	Katakri and PiTuKri criteria (Traficom NCSA-FI) for classified and cloud assessment
Data classification	Security Classification Decree: Restricted (IV) / Confidential (III) / Secret (II) / Top Secret (I)
Procurement route	Hansel Oy central purchasing body

Foreign jurisdiction exposure. Azure and Google regions live in Finland; TUVE network carries classified traffic outside them

Under the US CLOUD Act and FISA 702 a provider subject to US jurisdiction can face a lawful order for data it holds regardless of where that data sits. Residency is necessary but not sufficient; what matters is who holds the keys and who can be compelled.

6 Current state and provider landscape

Government cloud	Valtori (Government ICT Centre) state DCs + public-cloud brokerage; PiTuKri cloud security criteria; Digital Sovereignty Roadmap adopted Apr 2026; no branded sovereign cloud
Maturity	operational
Digital identity	Suomi.fi e-Identification (DVV)
Interconnection	FICIX Helsinki/Espoo/Oulu; C-Lion1 Helsinki-Rostock (damaged Nov 2024)

7 Migration path and cost

Phase	Scope	MW	CAPEX	Cumulative	Hybrid
1	Sovereign core	0.8	EUR 16 m	16%	no
2	Security and defense	1.7	EUR 42 m	56%	no
3	State record	0.7	EUR 18 m	74%	no
4	Elective	1.0	EUR 28 m	100%	yes

Phase 1 is the number that matters: **EUR 16 m** for 0.8 MW, 16% of total CAPEX. That is the floor below which no hybrid arrangement helps — and it is a small fraction of the full build.

8 Geography and threat notes

Very low seismic/flood; abundant land, cold climate, cheapest EU power (nuclear+wind+hydro); 1,340 km Russian border (NATO since 2023); Baltic Sea subsea sabotage risk; continental grid link only via Sweden/Estonia

9 Critical national data in scope

What the platform would *hold*, tiered by consequence of loss rather than by department. Tier 0 is the identity spine; tier 1 is the enforceable relationship between citizen and state. **0 of 15 record classes recorded.**

Tier	Record class	Register	Official description
0	Civil registry core	<i>not yet recorded</i>	
0	Facial biometric	<i>not yet recorded</i>	
0	Fingerprint biometric	<i>not yet recorded</i>	
0	Breeder document scans	<i>not yet recorded</i>	
0	Document issuance history	<i>not yet recorded</i>	
0	Digital identity credentials	<i>not yet recorded</i>	
0	Authentication audit log	<i>not yet recorded</i>	
0	Electoral roll entry	<i>not yet recorded</i>	
1	Tax	<i>not yet recorded</i>	
1	Benefits & pensions	<i>not yet recorded</i>	
1	Land & property registry	<i>not yet recorded</i>	
1	Judicial & criminal justice	<i>not yet recorded</i>	
1	Education	<i>not yet recorded</i>	
1	Business registry	<i>not yet recorded</i>	

Tier	Record class	Register	Official description
1	Vehicle & licensing	<i>not yet recorded</i>	

A blank row means this repository has not yet researched that register. It is not a statement that the country holds no such data. Tiers 2 (health records, imaging) and 3 (genomics, archives, video retention, geospatial) are out of scope: they hold most of the bytes, but Tiers 0 and 1 hold the sovereignty.